

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,C,D	B,C,D	B,C,D	A,C	A,B,C,D	A,C,D	A,B,C,D
SECTION-I (ii)	Q.	8	9	10	11	12	13	
	A.	C	A	D	C	A	D	
SECTION-II	Q.	1	2	3	4	5		
	A.	50.00	4.12 to 4.15	8.00	50.00	24.00		

PART-2 : CHEMISTRY

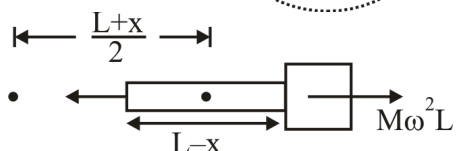
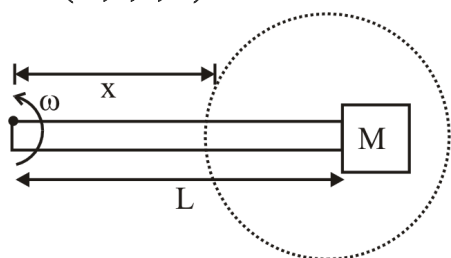
SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,B,C	A,B,C,D	A,B,C,D	B,C,D	A,B,D	B,D	A,B,D
SECTION-I (ii)	Q.	8	9	10	11	12	13	
	A.	A	D	A	D	C	B	
SECTION-II	Q.	1	2	3	4	5		
	A.	- 3.22	0.78	4.00	20.00	9.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7
	A.	A,C	B,C	A,D	B,C	A,B,C	B,C,D	A,C
SECTION-I (ii)	Q.	8	9	10	11	12	13	
	A.	C	D	C	C	A	C	
SECTION-II	Q.	1	2	3	4	5		
	A.	1.00	960.00	5.00	9.00	4.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

1. Ans (A,B,C,D)



$$T = M\omega^2 L + \mu(L-x)\omega^2 \left(\frac{L+x}{2} \right)$$

$$= M\omega^2 + \frac{\mu\omega^2}{2}(L^2 - x^2)$$

$$= \omega^2 \left[ML + \frac{\mu}{2}(L^2 - x^2) \right]$$

$$V = \sqrt{\frac{T}{\mu}} = f\lambda; \lambda = \frac{1}{f} \sqrt{\frac{T}{\mu}}$$

$$\lambda_2 = \lambda_1 \sqrt{\frac{T_2}{T_1}} = \lambda \sqrt{\frac{ML}{ML + \frac{\mu L^2}{2}}} = \lambda \sqrt{\frac{2M}{2M + \mu L}}$$

$$= \lambda \sqrt{\frac{2M}{2M + \mu L}} = \lambda \sqrt{\frac{6M}{8M}} = \sqrt{\frac{3}{4}} \lambda$$

$$V = \sqrt{\frac{T}{\mu}} = \omega \sqrt{\frac{ML + \frac{\mu}{2}(L^2 - x^2)}{\mu}}$$

$$= \omega \sqrt{\left(\frac{ML}{\mu} + \frac{L^2}{2} \right) - \frac{x^2}{2}} = \frac{\omega}{\sqrt{2}} \sqrt{\left(\frac{2ML}{\mu} + L^2 \right) - x^2}$$

$$= \frac{\omega}{\sqrt{2}} \sqrt{3L^2 + L^2 - x^2}$$

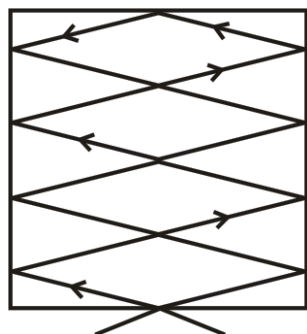
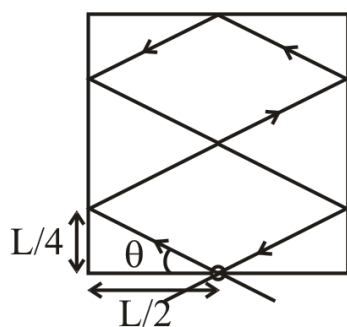
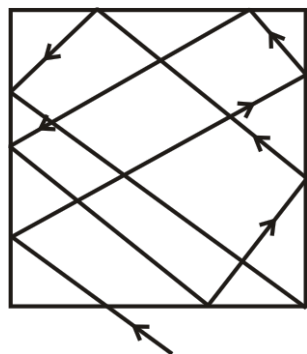
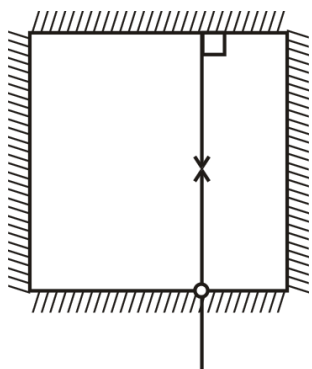
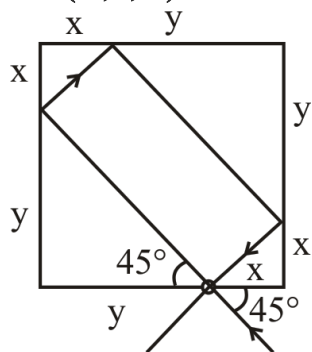
$$V = \frac{\omega}{\sqrt{2}} \sqrt{(2L)^2 - x^2} = \omega_0 \sqrt{A^2 - x^2}$$

$$\text{Time to move from } x = 0 \text{ to } x = \frac{A}{2}$$

$$\Rightarrow t = \frac{\text{time period}}{12} = \frac{2\pi}{\omega_0 \times 12}$$

$$= \frac{\pi}{6\left(\frac{\omega_0}{\sqrt{2}}\right)} = \frac{\pi}{3\sqrt{2}\omega}$$

2. Ans (B,C,D)



3. Ans (B,C,D)

$$\sqrt{D^2 + d^2} = D; \sqrt{D + 4d^2} = D$$

$$= D \left[\left(1 + \frac{d^2}{2D^2} - 1 \right) \right] = D \left[\left(1 + \frac{4d^2}{2D^2} - 1 \right) \right]$$

$$\Delta x_1 = \frac{d^2}{2D}; \Delta x_2 = \frac{2d^2}{D}$$

$$(A), (B) \text{ for } d = \sqrt{2\lambda D} \text{ and } 2\sqrt{\lambda D}$$

For A, B and C

$$AR = \overrightarrow{A} + \overrightarrow{A} + \overrightarrow{A} = 3A$$

$$I_R = 9A^2 = 9I_0$$

$$(C) \text{ For } d = \sqrt{\frac{\lambda D}{2}}$$

$$\Delta x_1 = \frac{\lambda}{4}, \Delta x_2 = \lambda$$

For A and C

$$\overrightarrow{A} + \overrightarrow{A} = \sqrt{5}A$$

$$I_R = 5I_0$$

For

$$\overrightarrow{B} + \overrightarrow{A} = \overrightarrow{A}$$

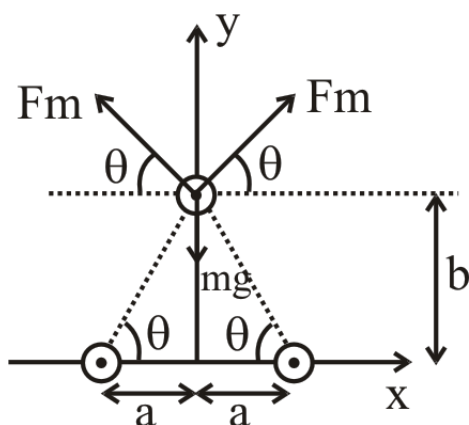
$$(D) \text{ For } d = \frac{\sqrt{\lambda D}}{2}, \Delta x_1 = \frac{\lambda}{8}; \Delta x_2 = \frac{\lambda}{2}$$

For A and C

$$\overrightarrow{A} + \overrightarrow{A} = \overrightarrow{A}$$

4.

Ans (A,C)



$$mg = 2Fm \sin \theta$$

$$\lambda l g = 2 \left[\frac{\mu_0 I i l}{2\pi \sqrt{a^2 + b^2}} \right] \left[\frac{b}{\sqrt{a^2 + b^2}} \right]$$

$$i = \frac{\pi \lambda g (a^2 + b^2)}{\mu_0 I b}$$

If wire is displaced by Δy

$$\Delta F = 2Fm \sin \theta - mg$$

$$\begin{aligned} &= \frac{\mu_0 I i l (b + \Delta y)}{\pi \{a^2 + (b + \Delta y)^2\}} - mg \\ &= \frac{\mu_0 I i l b}{\pi (a^2 + b^2)} \left\{ \frac{1 + \frac{\Delta y}{b}}{1 + \frac{2b\Delta y}{a^2 + b^2}} \right\} - mg \\ &= mg \left[1 + \Delta y \left(\frac{1}{b} - \frac{2b}{a^2 + b^2} \right) \right] - mg \\ &= \frac{mg(a^2 - b^2)}{b(a^2 + b^2)} \Delta y = - \frac{mg(b^2 - a^2)}{b(a^2 + b^2)} \Delta y \end{aligned}$$

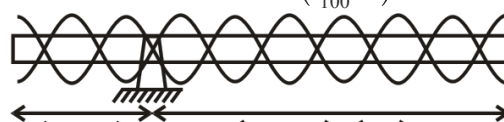
5.

Ans (A,B,C,D)

$$(A) V_{\text{rod}} = \sqrt{\frac{Y}{\rho}} = \sqrt{\frac{2.43 \times 10^9 \text{ N/m}^2}{3 \times 10^3 \text{ kg/m}^3}} = 900 \text{ m/s}$$

$$(B) V_{\text{gas}} = \sqrt{\frac{B}{\rho}} = \sqrt{\frac{9 \times 10^5 \text{ N/m}^2}{0.01 \times 10^3 \text{ kg/m}^3}} = 300 \text{ m/s}$$

$$(C) f_{\text{gas}} = \frac{V_{\text{gas}}}{\lambda_{\text{gas}}} = \frac{V_{\text{gas}}}{2\ell} = \frac{300 \text{ m/s}}{2 \left(\frac{3.33}{100} \text{ m} \right)} \approx 4500 \text{ Hz}$$



$$(D) \left(n + \frac{1}{2} \right) \left(m + \frac{1}{2} \right) \left(\frac{\lambda}{2} \right) = \frac{L}{2} = \frac{3L}{4}$$

$$\frac{2n+1}{2m} = \frac{1}{3} = \frac{3}{9} = \frac{5}{15} = \frac{7}{21} = \dots$$

$$f_{\text{Rod}} = f_{\text{gas}} = \frac{V_{\text{Rod}}}{\lambda_{\text{Rod}}}$$

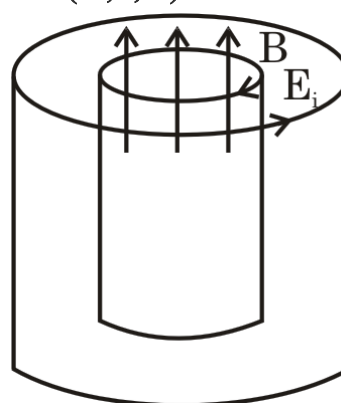
$$\lambda_{\text{Rod}} = \frac{V_{\text{Rod}}}{f_{\text{Rod}}} = \frac{900 \text{ m/s}}{4500 \text{ Hz}}$$

$$\frac{L}{2n+1} = 0.2 \text{ m}$$

$$2n+1 = \frac{1}{0.2} = 5 \Rightarrow 2^{\text{nd}} \text{ overtone}$$

6.

Ans (A,C,D)



$$\vec{s} = \frac{\vec{E} \times \vec{B}}{\mu_0}$$

$$E = \frac{r}{2} \frac{dB}{dt} = \mu_0 n i_0 \frac{\omega r}{2} \cos \omega t$$

$$B = \mu_0 n i_0 \sin \omega t$$

$$\vec{E} \times \vec{B} \text{ is inwards}$$

$$\frac{dU}{dt} = \frac{\mu_0 n^2 i_0^2 \omega r}{4} \sin 2\omega t \times 2\pi r \ell$$

PART-1 : PHYSICS

SECTION-I (ii)

11. Ans (C)

$$\frac{1}{f_{eq}} = \frac{1}{f_m} - \frac{2}{f_L}$$

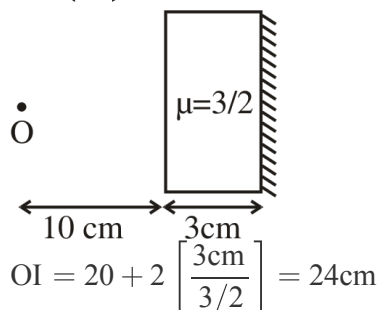
$$(A) \frac{1}{f_A} = \frac{1}{10} - \frac{2}{10} = \frac{-3}{10}$$

$$(B) \frac{1}{f_B} = \frac{1}{\infty} - \frac{2}{10} = \frac{-3}{5\text{cm}}$$

$$(C) \frac{1}{f_C} = \frac{1}{+15} - \frac{1}{5} = \frac{1-3}{15} = \frac{-2}{150}$$

$$(D) \frac{1}{f_D} = \frac{1}{+15} - \frac{1}{+20} = \frac{4-3}{60} = \frac{-2}{60}$$

12. Ans (A)



13. Ans (D)

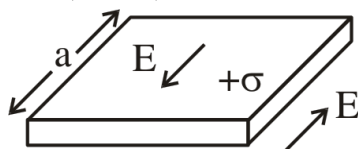
$$\frac{1}{f} = \frac{1}{f_1} = \frac{1}{f_2} = \frac{-1}{10} + \frac{1}{30} = \frac{-3+1}{30}$$

$$f = -15\text{cm}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (50.00)



$$\oint \vec{E} \cdot d\vec{\ell} = \frac{-d\phi}{dt}$$

$$E(2a) = \frac{B_0 ah}{\Delta t}$$

$$E = \frac{B_0 h}{2\Delta t}$$

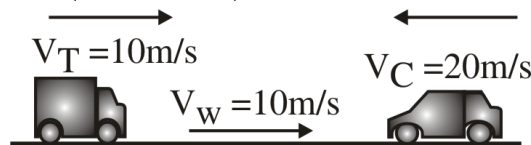
$$m(v-0) = 2qE\Delta t = 2\sigma(ab) \left[\frac{B_0 h}{2} \right]$$

$$v = \frac{\sigma ab h B_0}{m} = \frac{(10^{-2})(1)(1)(10^{-3})(5)}{1} \text{ m/s}$$

$$= 50 \times 10^{-6} \text{ m/s}$$

$$= 50 \mu\text{m/s}$$

2. Ans (4.12 to 4.15)



$$n = n_2 - n_1$$

$$= n \left[\left(\frac{V_s - V_\omega + V_T}{V_s - V_\omega - V_C} \right) \left(\frac{V_s + V_\omega + V_T}{V_s + V_\omega - V_C} \right) - 1 \right]$$

$$= 20 \left[\left(\frac{320}{290} \right) + \left(\frac{350}{320} \right) - 1 \right]$$

$$= 20 \times \left[\frac{6}{29} \right] \text{ Hz} = 4.14 \text{ Hz}$$

3. Ans (8.00)

$$v = \sqrt{\frac{2GM}{R^2}} h = \sqrt{2G\rho \left(\frac{4}{3} \pi R \right) h}$$

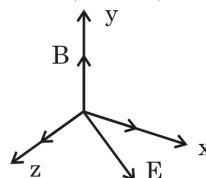
$$= \sqrt{\frac{2G}{r} \left(2\rho \times \frac{4}{3} \pi r^3 \right)}$$

$$\Rightarrow 2r^2 = Rh$$

$$r = \sqrt{\frac{Rh}{2}} = \sqrt{\frac{64 \times 10^5 \times 20}{2}}$$

$$= \sqrt{64 \times 10^6} \text{ m} = 8 \times 10^3 \text{ m} = 8 \text{ Km}$$

4. Ans (50.00)



$$\frac{\vec{k} \times \vec{E}}{\omega} = \vec{B}$$

$$\frac{\hat{k} \times 100\hat{i}}{3 \times 10^8} = \vec{B}$$

$$\vec{B} = \frac{1}{3} \times 10^{-6} \hat{j}$$

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B})$$

$$m\vec{a} = 10^{-6} \left(100\hat{i} + 9 \times 10^7 (\hat{i} + 2\hat{k}) \times \frac{1}{3} \times 10^{-6} \hat{j} \right)$$

$$\vec{a} = [100\hat{i} + 30\hat{k} - 60\hat{i}]$$

$$= 40\hat{i} + 30\hat{k} (\sin(2\pi z - \omega t))$$

$$a = 50 \text{ m/s}^2 \times \sin \omega t$$

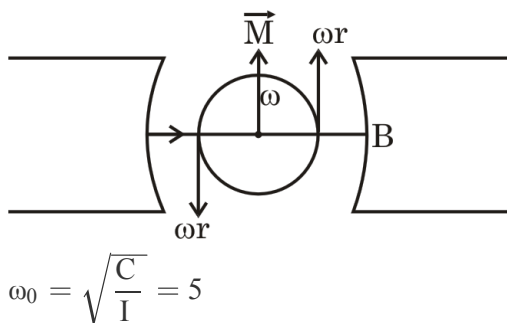
$$= 50 \times \sin(6\pi \times 10^8 \times \frac{5}{6} \times 10^{-9})$$

$$= 50 \text{ m/s}^2$$

$$\frac{\omega}{k} = 3 \times 10^8$$

$$\omega = 6\pi \times 10^8$$

5. Ans (24.00)



$$M = NiA$$

$$\tau = NiAB$$

$$\varepsilon = 2B\ell\omega r \times N = NBA\omega$$

$$i = \frac{\varepsilon}{R} = \frac{NBA\omega}{R}$$

$$\tau = \frac{-N^2 B^2 A^2 \omega}{R}$$

$$\tau_{\text{net}} = -C\theta - \frac{N^2 B^2 A^2 \omega}{R}$$

$$\frac{d^2\theta}{dt^2} + \frac{N^2 B^2 A^2}{IR} \frac{d\theta}{dt} + \frac{C}{I} \theta = 0$$

$$\gamma = \frac{N^2 B^2 A^2}{2IR}$$

$$\omega = 4 = \sqrt{\frac{C}{I} - \gamma^2}$$

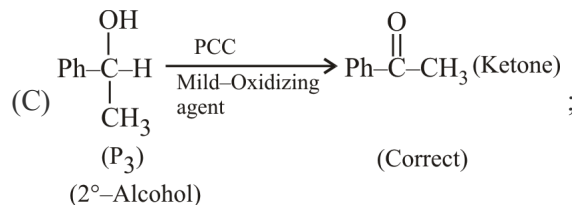
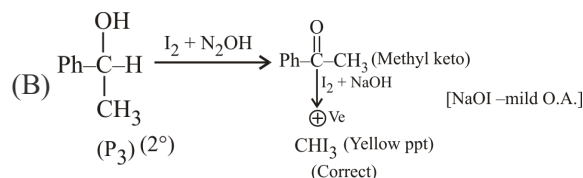
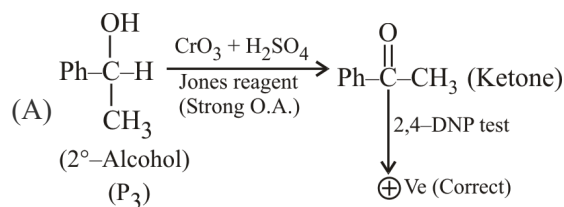
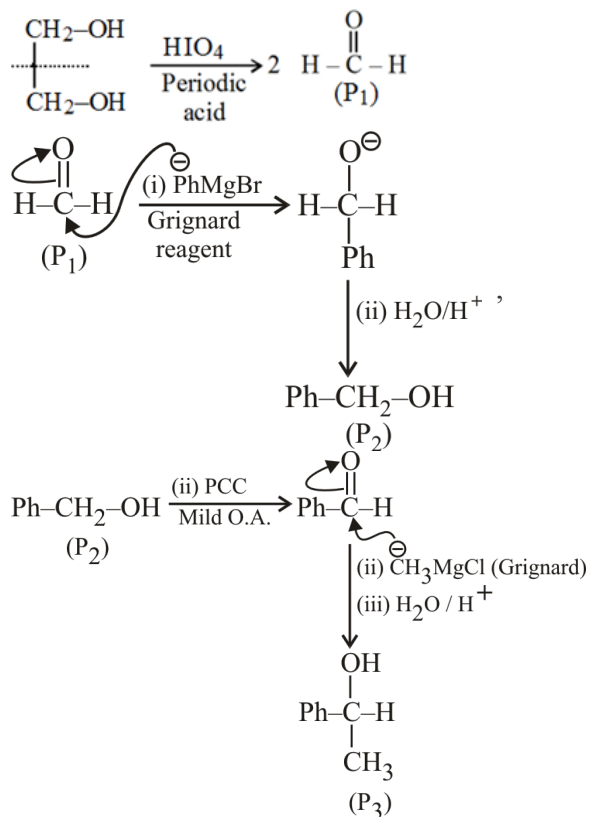
$$\gamma = 3 = \frac{4 \times 10^4 \times 0.09 \times 10^{-4}}{2 \times 2.5 \times R \times 10^{-3}}$$

$$R = 24 \Omega$$

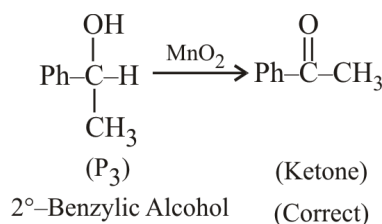
PART-2 : CHEMISTRY

SECTION-I (i)

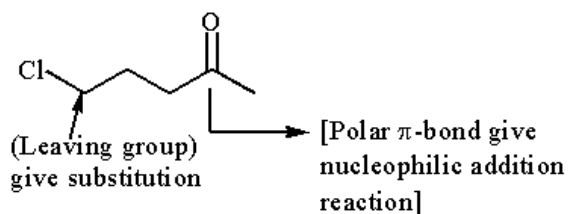
1. Ans (A,B,C)



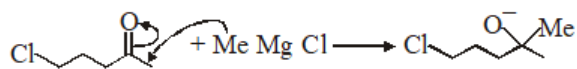
(D) MnO_2 converts 2° Allylic/Benzylic (–OH) into Ketone



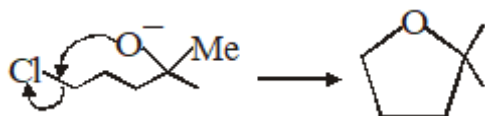
2. Ans (A,B,C,D)



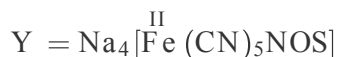
(i) Grignard prefer to give nucleophilic addition on polar π -bond and from anion intermediate.



(ii) In next step anion give intramolecular nucleophilic substitution reaction & form 5 membered ring.



6. Ans (B,D)



7. Ans (A,B,D)

After mixing $[\text{H}^+] = 10^{-3} \text{ M}$

$$5 \times 10^{-6} = \frac{[\text{A}^-][\text{H}^+]}{[\text{HA}]} = \frac{C\alpha}{C(1-\alpha)} \times 0.1$$

$$\frac{\alpha}{1-\alpha} = 5 \times 10^{-5} \Rightarrow \alpha \approx 5 \times 10^{-5}$$

$$[\text{A}^-] = C\alpha = 5 \times 10^{-6} \text{ M}$$

PART-2 : CHEMISTRY

SECTION-II

1. Ans (- 3.22)

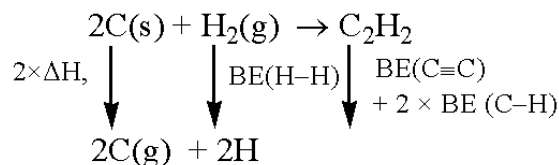
$$\Delta_r G = RT \ln$$

$$\frac{p_{\text{H}_2\text{O}}}{V \cdot p_{\text{H}_2\text{O}}} = RT \ln(0.01)$$

$$= \frac{2 \times 350}{1000} \times 2.303(-2)$$

$$= -3.22 \text{ kcal/mole}$$

2. Ans (0.78)



$$\Delta H_{\text{rxn}} = \Delta_f H(\text{C}_2\text{H}_2) = [2 \times \Delta H_f + \text{BE(H-H)}] - [\text{BE(C}\equiv\text{C)} + 2 \times \text{BE(C-H)}]$$

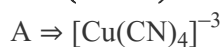
$$\Rightarrow 220 = (1400 + 300) - (\text{BE(C}\equiv\text{C)} + 2 \times 350)$$

$$\Rightarrow \text{BE(C}\equiv\text{C)} = 780 \text{ kJ/mol}$$

3. Ans (4.00)



4. Ans (20.00)



5. Ans (9.00)

No. of atoms present in one hcp unit cell

$$12 \text{ on corners} \quad 12 =$$

$$\left(\frac{1}{6} \text{ contribution} \right) = 12 \times \frac{1}{6}$$

$$2 \text{ on 2 face centres} \quad 2 =$$

$$\left(\frac{1}{2} \text{ contribution} \right) = 2 \times \frac{1}{2}$$

$$3 \text{ on mid layer} \quad 3 (100\% \text{ contributing}) = 3$$

$$\text{Total} = 2 + 1 + 3 = 6$$

No. of tetra hedral voids = 12 voids

No. of octa hedral voids = 6 voids

12 migram Mg $\Rightarrow$

$$\frac{12}{24} \times 10^{-3} = \frac{1}{2} \times 10^{-3} \text{ mole of Mg}$$

Number of Mg atoms =

$$\frac{1}{2} \times 10^{-3} \times 6 \times 10^{23} = 3 \times 10^{20} \text{ atoms}$$

No. of hcp unit cell if one unit cell occupy 6 atoms

$$= \frac{1}{2} \times 10^{20} \text{ HCP unit cell}$$

1 unit cell has 18 (12 (TV) + 6 (OV) voids

$$\frac{1}{2} \times 10^{20} \text{ hcp unit cell has} = 9 \times 10^{20} \text{ voids}$$

$$x \times 10^{20} \text{ voids} \Rightarrow x \text{ value will be} = 9$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. Ans (A,C)

Multiplying the first DE by gh, then second by fh and the third by fg, and adding the equations gives

$$(fgh)' = 6(fgh)^2 + 6$$

$$\text{Let } f(x)g(x)h(x) = k(x)$$

$$\text{We have } k'(x) = 6(k(x))^2 + 6$$

Integrating and using $k(0) = 1$ gives

$$k(x) = \tan\left(6x + \frac{\pi}{4}\right)$$

$$\text{Now 1st DE} \Rightarrow \frac{f'(x)}{f(x)} = 2k(x) + \frac{1}{k(x)}$$

$$\Rightarrow \frac{f'(x)}{f(x)} = 2 \tan\left(6x + \frac{\pi}{4}\right) + \cot\left(6x + \frac{\pi}{4}\right)$$

Integrating and using $f(0) = 1$ gives option A and C

2. Ans (B,C)

Put $x = y + 1$

$$\underbrace{f(y+1) - f(y)}_{\text{Differentiable}} = f'\left(\frac{x+y}{2}\right) \quad \dots(1)$$

$\therefore f(x)$ is twice differentiable $\forall x \in \mathbb{R}$

Differentiating the given expression w.r.t. x

$$f'(x) = f'\left(\frac{x+y}{2}\right) + \left(\frac{x-y}{2}\right) f''\left(\frac{x+y}{2}\right) \quad \dots(2)$$

Differentiating the given expression w.r.t. y

$$-f'(y) = -f'\left(\frac{x+y}{2}\right) + \left(\frac{x-y}{2}\right) f''\left(\frac{x+y}{2}\right) \quad \dots(3)$$

Eliminating $f''\left(\frac{x+y}{2}\right)$ from (2) and (3)

$$\frac{f'(x) + f'(y)}{2} = f'\left(\frac{x+y}{2}\right) \quad \dots(4)$$

Differentiating (4) w.r.t. x and y separating

$$\frac{1}{2} f''\left(\frac{x+y}{2}\right) = \frac{f''(x)}{2} \quad \text{and}$$

$$\frac{1}{2} f''\left(\frac{x+y}{2}\right) = \frac{f''(y)}{2}$$

$$\therefore f''(x) = f''(y) = \text{constant}$$

$$\therefore f(x) = ax^2 + bx + c$$

$$f(0) = f(1) = 0 \Rightarrow c = 0, a + b = 0$$

$$\therefore f(x) = a(x^2 - x) = ax(x-1)$$

$$\text{and } f(2) = 2a > 0 \Rightarrow a > 0$$

3. Ans (A,D)

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}; ad - bc \neq 0$$

$$a + b + c + d = 5, a_{ij} \in \{0, 1, 2, 3, 4, 5\}$$

(A) (i) Exactly one zero

$$\text{Number of matrices} = \frac{4!}{2!} + \frac{4!}{2!} = 24$$

(ii) Exactly 2 zeros

$$\text{Number of matrices} = 2 \times {}^{5-1}C_{2-1} = 8$$

$$\therefore \text{Total matrices} = 32$$

(B) Number of symmetric matrices =

$$\sum_{b=0}^2 (6-2b) = 12$$

(C) Det $A = ad - bc$ is odd

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \text{ if } A = A^T \text{ then } b = c$$

$$\therefore \text{Det } A = ad - b^2, a + d = 5 - 2b$$

(i) $b = 0, a + d = 5$ not possible

(ii) $b = 1, a + d = 3$ 4 cases

(iii) $b = 2, a + d = 1$ not possible

4. Ans (B,C)

$$|\vec{a}| = 1, |\vec{b}| = 4, \vec{a} \cdot \vec{b} = 2 \text{ \& } \vec{c} + 3\vec{b} = 2\vec{a} \times \vec{b}$$

$$\therefore \vec{a} \cdot \vec{b} = 2$$

$$\Rightarrow |\vec{a}| |\vec{b}| \cos \theta = 2$$

$$\Rightarrow \cos \theta = \frac{2}{|\vec{a}| |\vec{b}|} = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$$

$$\text{Again } |\vec{c} + 3\vec{b}|^2 = |2\vec{a} \times \vec{b}|^2$$

$$\Rightarrow |\vec{c}|^2 + 9|\vec{b}|^2 + 2\vec{c} \cdot 3\vec{b} = 4|\vec{a}|^2 |\vec{b}|^2 \sin^2 \theta$$

$$\Rightarrow |\vec{c}|^2 + 144 + 6\vec{b} \cdot \vec{c} = 48$$

$$\text{Also, } \vec{c} = 2\vec{a} \times \vec{b} - 3\vec{b}$$

$$\Rightarrow \vec{b} \cdot \vec{c} = \vec{b} \cdot (2\vec{a} \times \vec{b}) - 3|\vec{b}|^2$$

$$\Rightarrow \vec{b} \cdot \vec{c} = 0 - 3 \times 16 = -48$$

$$\therefore |\vec{c}|^2 + 96 - 6 \times 48 = 0$$

$$\Rightarrow |\vec{c}|^2 = 192$$

$$\Rightarrow |\vec{c}| = 8\sqrt{3}$$

$$\vec{b} \cdot \vec{c} = |\vec{b}| |\vec{c}| \cos \alpha \Rightarrow -48 = 4 \times 8\sqrt{3} \cos \alpha$$

$$\Rightarrow \cos \alpha = -\frac{6}{4\sqrt{3}} = -\frac{\sqrt{3}}{2}$$

$$\Rightarrow \alpha = \frac{5\pi}{6}$$

5. Ans (A,B,C)

Take base as the x-y plane with one vertex as origin. A, B be vertices along the x, y-axes respectively

$$\vec{OA} = 2\hat{i}, \vec{OB} = 2\hat{j}, \vec{OC} = \hat{i} + \hat{j} + 2\hat{k}, \text{ where}$$

C is the apex

$$\therefore \vec{OA} \times \vec{OB} = 4\hat{k}; \vec{OA} \times \vec{OC} = 2\hat{k} - 4\hat{j};$$

$$\vec{OB} \times \vec{OC} = -2\hat{k} + 4\hat{i}$$

$$\text{Volume is } \frac{1}{3} \times \text{parallelepiped} = \frac{8}{3}$$

Tetrahedron with half of pyramid's base and the pyramid's apex as volume = $\frac{4}{3}$

6. Ans (B,C,D)

$$\because [\vec{a} \ \vec{b} \ \vec{c}] = 2 \text{ and } 3(\vec{a} + \vec{b}) \cdot \{(\vec{b} + \vec{c}) \times (\vec{c} + \vec{a})\}$$

$$= 3(\vec{a} + \vec{b}) \cdot \{\vec{b} \times \vec{c} + \vec{b} \times \vec{a} + 0 + \vec{c} + \vec{a}\}$$

$$= 3\{2[\vec{a} \ \vec{b} \ \vec{c}]\} = 6[\vec{a} \ \vec{b} \ \vec{c}] = 12$$

$$[\vec{a} \times \vec{b} \ \vec{b} \times \vec{c} \ \vec{c} \times \vec{a}] = [\vec{a} \ \vec{b} \ \vec{c}]^2 = 4$$

$$\Rightarrow (\vec{a} - 2\vec{b}) \cdot \{(\vec{b} - 2\vec{c}) \times (\vec{c} - 2\vec{a})\}$$

$$= (\vec{a} - 2\vec{b}) \cdot \{\vec{b} \times \vec{c} - 2\vec{b} \times \vec{a} + 4\vec{c} \times \vec{a}\}$$

$$= [\vec{a} \ \vec{b} \ \vec{c}] - 8[\vec{b} \ \vec{c} \ \vec{a}] = -7[\vec{a} \ \vec{b} \ \vec{c}]$$

$$(3\vec{b} - \vec{c}) \times (3\vec{c} + \vec{a})$$

$$= 9\vec{b} \times \vec{c} - 3\vec{b} \times \vec{a} + \vec{c} \times \vec{a}$$

$$(3\vec{a} - \vec{b}) \cdot \{3\vec{c} + \vec{a}\}$$

$$= 27[\vec{a} \ \vec{b} \ \vec{c}] - [\vec{b} \ \vec{c} \ \vec{a}] = 26[\vec{a} \ \vec{b} \ \vec{c}]$$

7. Ans (A,C)

$$(1 + \sqrt{5})^n = \alpha_n + \beta_n \sqrt{5} \quad \dots(1)$$

$$(1 - \sqrt{5})^n = \alpha_n - \beta_n \sqrt{5} \quad \dots(2)$$

equation (1) and (2)

$$\Rightarrow (-4)^n = \alpha_n^2 - 5\beta_n^2$$

$$\Rightarrow (-1)^n 2^{2n} = \left(\frac{\alpha_n}{2^n}\right)^2 - 5\left(\frac{\beta_n}{2^n}\right)^2$$

$$\Rightarrow \alpha_{n+1} = \alpha_n + 5\beta_n \quad \dots(3)$$

$$\Rightarrow \beta_{n+1} = \beta_n + \alpha_n \quad \dots(4)$$

from equation (3) and (4) we have

$$\Rightarrow \alpha_{n+1} - \beta_{n+1} = \alpha_n + 5\beta_n - \beta_n - \alpha_n$$

Now put $n \rightarrow n+1$ in equation (1)

$$\Rightarrow \alpha_{n+1} + \beta_{n+1} \sqrt{5} = (1 + \sqrt{5})^{n+1}$$

$$(1 + \sqrt{5})(1 + \sqrt{5})^n = (1 + \sqrt{5})(\alpha_n + \beta_n \sqrt{5})$$

$$\Rightarrow \alpha_{n+1} + \beta_{n+1} \sqrt{5}$$

$$= \alpha_n + \beta_n \sqrt{5} + \sqrt{5} \alpha_n + 5\beta_n$$

$$\Rightarrow \frac{\alpha_{n+1} - \beta_{n+1}}{3} = \frac{4}{3} \beta_n, \text{ now check the option(s)}$$

$$\lim_{\theta \rightarrow 0} \frac{\tan^{-1}\left(\frac{4}{3}\beta_n \theta\right)}{\sin(\beta_n \theta)} = \frac{4}{3}$$

PART-3 : MATHEMATICS

SECTION-I (ii)

8. Ans (C)

$$(I) \ y = \sin x + \cos x \quad (II) \ y = \sin 2x$$

$$y' = \cos x - \sin x \quad y' = 2\cos 2x$$

$$y \uparrow \left(0, \frac{\pi}{4}\right) \quad y \uparrow \left(0, \frac{\pi}{4}\right)$$

$$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right) \quad y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$(III) \ y = \sin^5 x + \cos^5 x - 1$$

$$y' = 5\sin^4 x \cos x - 5\cos^4 x \sin x$$

$$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$$

$$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$$

$$y \downarrow \left(0, \frac{\pi}{4}\right)$$

$$y = x^e \quad y' = x^x(1 + \ln x)$$

$$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$$

$$y \downarrow \left(0, \frac{1}{e}\right)$$

9. Ans (D)

(I) $y = \sin x + \cos x$ (II) $y = \sin 2x$

$y' = \cos x - \sin x$ $y' = 2\cos 2x$

$y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$

$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$

$y' = 5\sin^4 x \cos x - 5\cos^4 x \sin x$

$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$

$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{\pi}{4}\right)$

$y = x^e$ $y' = x^x(1 + \ln x)$

$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{1}{e}\right)$

10. Ans (C)

(I) $y = \sin x + \cos x$ (II) $y = \sin 2x$

$y' = \cos x - \sin x$ $y' = 2\cos 2x$

$y \uparrow \left(0, \frac{\pi}{4}\right)$ $y \uparrow \left(0, \frac{\pi}{4}\right)$

$y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ $y \downarrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

(III) $y = \sin^5 x + \cos^5 x - 1$

$y' = 5\sin^4 x \cos x - 5\cos^4 x \sin x$

$= 5\sin x \cos x (\sin^3 x - \cos^3 x)$

$y \uparrow \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{\pi}{4}\right)$

$y = x^e$ $y' = x^x(1 + \ln x)$

$y \uparrow \left(\frac{1}{e}, \frac{\pi}{2}\right)$

$y \downarrow \left(0, \frac{1}{e}\right)$

11. Ans (C)

(I) $\int_0^{\pi/2} \sin^8 x \, dx = \frac{35\pi}{2^8} = \frac{A}{B-1} = 5$

(II) $\frac{\int_0^8 [x] \, dx}{\int_0^8 \{x\} \, dx} = \frac{1+2+\dots+7}{8 \cdot \frac{1}{2}} = 7$

(III) $\lim_{t \rightarrow 0} \frac{\int_0^t \frac{x^2 \ln(1+x)}{x^7+3} \, dx}{t^4} \Rightarrow \frac{t^2 \ln(1+x)}{(t^7+3)4t^3}$
 $= \frac{1}{12} \Rightarrow \frac{3}{4t} = 9$

(IV) $\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \sqrt{\frac{n}{r}} = \int_0^4 \frac{1}{\sqrt{x}} \, dx$

$2\sqrt{x} \Big|_0^4 = 4$

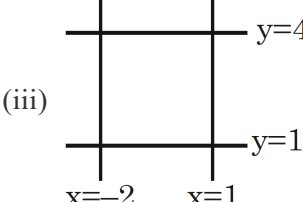
(i) $\frac{dy}{dx} = 1 + \cos 2x \Big|_{x=\frac{\pi}{2}} = 0$

$m+7=7$

(ii) $y - \left(\frac{7}{2}\right) = \frac{k-2}{3} \left(x - \frac{k+2}{2}\right)$

$2 = \frac{(k-2)(k+2)}{6}$

$k^2 = 16 \Rightarrow k = 4$

(iii)  area = 9

(iv) $A = \frac{\ell^2 \sqrt{3}}{4}$

$\frac{dA}{d\ell} = \frac{\ell \sqrt{3}}{2} \Big|_{\frac{10}{\sqrt{3}}} = 5$

12. Ans (A)

$$(I) \int_0^{\pi/2} \sin^8 x \, dx = \frac{35\pi}{2^8} = \frac{A}{B-1} = 5$$

$$(II) \frac{\int_0^8 [x] \, dx}{\int_0^8 \{x\} \, dx} = \frac{1+2+\dots+7}{8 \cdot \frac{1}{2}} = 7$$

$$(III) \lim_{t \rightarrow 0} \frac{\int_0^t \frac{x^2 \ln(1+x)}{x^7+3} \, dx}{t^4} \Rightarrow \frac{t^2 \ln(1+x)}{(t^7+3)4t^3}$$

$$= \frac{1}{12} \Rightarrow \frac{3}{4\ell} = 9$$

$$(IV) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \sqrt{\frac{n}{r}} = \int_0^4 \frac{1}{\sqrt{x}} \, dx$$

$$2\sqrt{x} \Big|_0^4 = 4$$

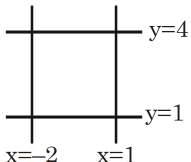
$$(i) \frac{dy}{dx} = 1 + \cos 2x \Big|_{x=\frac{\pi}{2}} = 0$$

$$m+7=7$$

$$(ii) y - \left(\frac{7}{2}\right) = \frac{k-2}{3} \left(x - \frac{k+2}{2}\right)$$

$$2 = \frac{(k-2)(k+2)}{6}$$

$$k^2 = 16 \Rightarrow k = 4$$

(iii)  area = 9

$$(iv) A = \frac{\ell^2 \sqrt{3}}{4}$$

$$\frac{dA}{d\ell} = \frac{\ell \sqrt{3}}{2} \Big|_{\frac{10}{\sqrt{3}}} = 5$$

13. Ans (C)

$$(I) \int_0^{\pi/2} \sin^8 x \, dx = \frac{35\pi}{2^8} = \frac{A}{B-1} = 5$$

$$(II) \frac{\int_0^8 [x] \, dx}{\int_0^8 \{x\} \, dx} = \frac{1+2+\dots+7}{8 \cdot \frac{1}{2}} = 7$$

$$(III) \lim_{t \rightarrow 0} \frac{\int_0^t \frac{x^2 \ln(1+x)}{x^7+3} \, dx}{t^4} \Rightarrow \frac{t^2 \ln(1+x)}{(t^7+3)4t^3}$$

$$= \frac{1}{12} \Rightarrow \frac{3}{4\ell} = 9$$

$$(IV) \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^{4n} \sqrt{\frac{n}{r}} = \int_0^4 \frac{1}{\sqrt{x}} \, dx$$

$$2\sqrt{x} \Big|_0^4 = 4$$

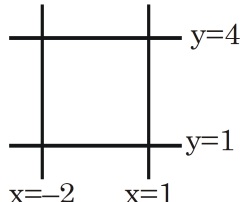
$$(i) \frac{dy}{dx} = 1 + \cos 2x \Big|_{x=\frac{\pi}{2}} = 0$$

$$m+7=7$$

$$(ii) y - \left(\frac{7}{2}\right) = \frac{k-2}{3} \left(x - \frac{k+2}{2}\right)$$

$$2 = \frac{(k-2)(k+2)}{6}$$

$$k^2 = 16 \Rightarrow k = 4$$

(iii)  area = 9

$$(iv) A = \frac{\ell^2 \sqrt{3}}{4}$$

$$\frac{dA}{d\ell} = \frac{\ell \sqrt{3}}{2} \Big|_{\frac{10}{\sqrt{3}}} = 5$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (1.00)

$$2 = (a_{11} + a_{12} + a_{13} + a_{14}) + (a_{41} + a_{42} + a_{43} + a_{44}) + (a_{11} + a_{22} + a_{33} + a_{44}) + (a_{14} + a_{23} + a_{32})$$

$$- (a_{12} + a_{22} + a_{32} + a_{42}) - (a_{13} + a_{23} + a_{33} + a_{43})$$

$$= 2(a_{11} + a_{14} + a_{41} + a_{44})$$

2. Ans (960.00)

$$\begin{matrix} & A & & B \\ \begin{bmatrix} 4\lambda_1 + 1 & 4\lambda_2 + 1 \\ 4\lambda_3 + 1 & 4\lambda_4 + 1 \end{bmatrix} & \times & \begin{bmatrix} - & - & - \\ - & - & - \end{bmatrix} \\ = & \begin{bmatrix} 4\lambda & 4\lambda \\ 4\lambda & 4\lambda \end{bmatrix} & &
 \end{matrix}$$

$\Rightarrow B$ can be of the form

$$\begin{bmatrix} 4\lambda_5 - 1 & 4\lambda_6 - 1 \\ 4\lambda_7 + 1 & 4\lambda_8 + 1 \end{bmatrix} \text{ Interchange possible}$$

$$\Rightarrow 4 \times {}^5P_2 \times {}^4P_2 = 4 \times 20 \times 12 = 960$$

3. Ans (5.00)

$$A = \begin{bmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{bmatrix}$$

$$\Rightarrow B = \frac{1}{3}A^{-1} = \frac{1}{3} \begin{bmatrix} 9 & -36 & 30 \\ -36 & 192 & -180 \\ 30 & -120 & 180 \end{bmatrix}$$

$$\Rightarrow B = \begin{bmatrix} 3 & -12 & 10 \\ -12 & 64 & -60 \\ 10 & -60 & 60 \end{bmatrix}$$

4. Ans (9.00)

Assume O to be origin and let circum radius be R and

$$\overrightarrow{OA} = \vec{a}, \overrightarrow{OB} = \vec{b}, \overrightarrow{OC} = \vec{c}$$

$$|\overrightarrow{AB}|^2 = |\vec{a} - \vec{b}|^2 = 2R^2 - 2\vec{a} \cdot \vec{b}$$

$$|\overrightarrow{AC}|^2 = |\vec{a} - \vec{c}|^2 = 2R^2 - 2\vec{a} \cdot \vec{c}$$

$$\overrightarrow{AO} \cdot \overrightarrow{BC} = -\vec{a} \cdot (\vec{c} - \vec{b}) = \vec{a} \cdot \vec{b} - \vec{a} \cdot \vec{c}$$

$$= \left(R^2 - \frac{1}{2} |\overrightarrow{AB}|^2 \right) - \left(R^2 - \frac{1}{2} |\overrightarrow{AC}|^2 \right)$$

$$= \frac{1}{2} \left(|\overrightarrow{AC}|^2 - |\overrightarrow{AB}|^2 \right)$$

$$\Rightarrow 2 \cdot \overrightarrow{AO} \cdot \overrightarrow{BC} = |\overrightarrow{AC}|^2 - |\overrightarrow{AB}|^2 = 9$$

5. Ans (4.00)

$$a_n = \frac{(2n+1)^9 C_{n+1}}{9} \text{ and}$$

$$b_n = \frac{(3n+1)^8 C_{n+1}}{8}$$

a_n is max when $n+1=5$ and b_n is min when $n=0$

$$\Rightarrow p = 4 \text{ and } q = 0$$

$$p + q = 4$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE (Main)

UNIT TEST

18-09-2022

JEE(Main + Advanced) : ENTHUSIAST COURSE [PHASE : S, I(A) & LIVE-I(A)]

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	A	A	A	A	D	A	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	D	C	A	B	B	B	C	B	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	30	7	4	16	100	8	3	30	2	100

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	B	B	A	A	C	D	D	C	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	C	B	B	B	C	A	C	D	A	A
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	6	4	476	3360	45	6	321	6	213	321

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	D	A	B	A	D	A	D	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	C	B	A	C	C	A	A	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	14	12	8	44	15	9	12	120	4	1

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. **Ans (B)**

$$1 \sin i = \mu \sin 37^\circ$$

$$\tan i = \mu$$

$$\Rightarrow \cos i = \sin 37^\circ$$

$$\Rightarrow i = 53^\circ$$

$$\mu = 4/3$$

2. **Ans (C)**

$$\text{At path difference } \frac{\lambda}{6}, \text{ phase difference is } \frac{\pi}{3}$$

$$I = I_0 + I_0 + 2I_0 \cos \frac{\pi}{3} = 3I_0 \quad I_{\max} = 4I_0$$

$$\text{So the required ratio is } \frac{3I_0}{4I_0} = 0.75$$

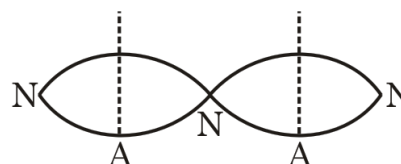
3. **Ans (A)**

Pressure change will be minimum at both ends.

4. **Ans (A)**

3 = Nodes, 2 = antinodes

distance = 1.21 Å



$$d = 1.21 \text{ Å}$$

No. of loops = 2

$$\lambda = 1.21 \text{ Å}$$

5. **Ans (A)**

Avoiding end correction, the length of closed organ pipe is

$$\ell_2 = \frac{\lambda_1}{4} \text{ or } \lambda_1 = 4\ell_1$$

The length of open organ pipe is

$$\ell_2 = \frac{\lambda_2}{2} \text{ or } \lambda_2 = 2\ell_2$$

Here $n_1 = n_2$

$$\Rightarrow \frac{v}{\lambda_1} = \frac{v}{\lambda_2} \text{ or } \frac{v}{4\ell_1} = \frac{v}{4\ell_2}$$

Therefore $\ell_1 : \ell_2 = 1 : 2$

6. **Ans (A)**

The apparent frequency heard by traveller moving towards stationary train

$$n' = n \left(\frac{v + v_0}{v + v_s} \right) = n \left(\frac{v + v_0}{v} \right) \quad (\because v_s = 0)$$

Where v = speed of sound.

v_0 = speed of traveller

7. **Ans (D)**

$$\because \omega = \frac{2\pi}{0.01}, k = \frac{2\pi}{0.3}$$

$$\frac{\omega}{v} = k \Rightarrow v = \frac{\omega}{k} = \frac{2\pi}{0.01} \times \frac{0.3}{2\pi}$$

$v = 30$ units

8. **Ans (A)**

$$\frac{T}{4} = 0.17 \text{ sec}$$

$$T = 0.17 \times 4 = 0.68 \text{ sec.}$$

$$f = \frac{1}{T} = \frac{1}{0.68} = \frac{100}{68} = 1.47 \text{ Hz}$$

10. **Ans (A)**

$$\frac{6 \times 10^3}{4\pi \times 10^4} = \frac{1}{2} C \epsilon_0 C^2 B_0^2$$

$$B_0^2 = \frac{6 \times 10 \times 9 \times 10 \times 2}{10^4 \times 27 \times 10^{24}}$$

11. **Ans (D)**

$$a = A \sin \omega t_0 \dots\dots(1)$$

$$b = A \sin 2\omega t_0 \dots\dots(2)$$

$$c = A \sin 3\omega t_0 \dots\dots(3)$$

Adding (1) and (3)

$$a + c = 2A \sin 2\omega t_0 \cos \omega t_0$$

From (2)

$$2b = 2A \sin 2\omega t_0$$

$$\omega = \frac{1}{t_0} \cos^{-1} \left(\frac{a+c}{2b} \right)$$

12. **Ans (C)**

$$\tau = \frac{mgR\theta}{2} = I\alpha$$

$$\alpha = \frac{mgR}{2I} \theta$$

$$\omega = \sqrt{\frac{mgR}{2I}} = \sqrt{\frac{mgR}{\frac{4}{3}mR^2}} = \sqrt{\frac{3g}{4R}}$$

$$f = \frac{\omega}{2\pi}$$

$$f = \frac{1}{4\pi} \sqrt{\frac{3g}{R}}$$

13. **Ans (A)**

$$v_y = \sqrt{\frac{T_y}{\mu_y}}$$

$$T_y = \left\{ \int_0^y \mu_0 e^y dy \right\} g$$

$$T_y = \mu_0 (e^y - 1) \cdot g$$

$$v_y = \sqrt{g - \frac{g}{e^y}}$$

$$v_y^2 = g(1 - e^{-y})$$

14. **Ans (B)**

For interference at A : S_2 is behind of S_1 by a distance

of $100\lambda + \frac{\lambda}{4}$ (equal to phase difference $\frac{\pi}{2}$). Further

S_2 lags S_1 by $\frac{\pi}{2}$. Hence the waves from S_1 and

S_2 interference at A with a phase difference of 200.5

$$\pi + 0.5\pi = 201\pi$$

Hence the net amplitude at A is $2a - a = a$

For interference at B : S_2 is ahead of S_1 by a distance

of $100\pi + \frac{\lambda}{4}$ (equal to phase difference $\frac{\pi}{2}$). Further

S_2 lags S_1 by $\frac{\pi}{2}$.

Hence waves from S_1 and S_2 interfere at B with a

phase difference of 200π

Hence the net amplitude at A is $2a + a = 3a$

15. **Ans (B)**

$$V_L = \frac{E}{2} e^{-t/\tau}$$

$$V_L = \frac{E}{4}$$

$$t = \tau \ln 2$$

$$t = \frac{2L}{R} \ln 2 = \frac{L}{R} \ln 4$$

16. **Ans (B)**

$$\text{Reading } V_2 = \sqrt{V_R^2 + V_C^2} = I\sqrt{R^2 + X_C^2}$$

$$8\sqrt{2} = I\sqrt{4^2 + 4^2} = 14\sqrt{2} \quad \therefore I = 2\text{Amp}$$

18. **Ans (B)**

Field induction $B_e = 0.36$ gauss

Angle of dip $\theta = 60^\circ$

The horizontal component is

$$H = B_e \cos \theta = 0.36 \times \cos 60^\circ$$

$$= 0.36 \times \frac{1}{2} = 0.18 \text{ gauss}$$

The vertical component is

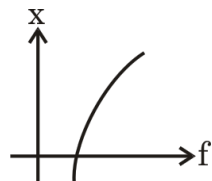
$$B_e \sin 60^\circ = 0.36 \times \sin 60^\circ$$

$$= 0.36 \times \frac{\sqrt{3}}{2} = 0.18 \times \sqrt{3} \text{ gauss}$$

19. **Ans (A)**

$$X = \left(\omega L - \frac{1}{\omega C} \right).$$

Hence the graph of X vs. f is



20. **Ans (C)**

$$R_1 = \frac{\rho \ell}{A}, R_2 = \frac{\rho \times 2\ell}{\frac{A}{2}} = 4R_1,$$

$$R_2 = \frac{\rho \ell}{2A} = \frac{R_1}{4}$$

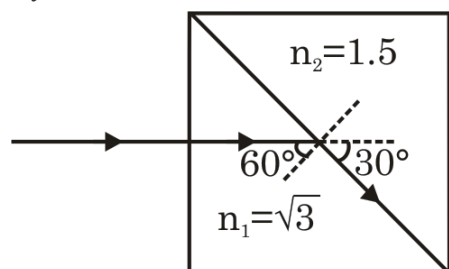
PART-1 : PHYSICS

SECTION-II

1. **Ans (30)**

Angle of incident at the boundary = 60°

By snell's law



2. **Ans (7)**

$$\frac{GM}{(2R)^2} = \frac{GM'}{R^2}$$

$$\frac{M}{4} = M'$$

$$\frac{4}{3}\pi R^3 \rho_1 + \frac{4}{3}\pi (8R^3 - R^3) \rho_2$$

$$= 4 \left(\frac{4}{3} \cdot \pi R^3 \cdot \rho_1 \right)$$

$$\rho_1 + 7\rho_2 = 4\rho_1$$

$$\frac{\rho_1}{\rho_2} = \frac{7}{3}$$

3. **Ans (4)**

$$F = -\frac{dU}{dx} = \frac{2a}{x^3} - \frac{b}{x^2} = \frac{1}{x^2} \left[\frac{2a}{x} - b \right]$$

$$F = 0 \text{ at } x = \frac{2a}{b}$$

$$\text{Let } F = \frac{2a}{\left(\frac{2a}{b} + y \right)^3} - \frac{b}{\left(\frac{2a}{b} + y \right)}$$

$$= \frac{2b^3a}{8a^3} \left(1 - \frac{3by}{2a} \right) - \frac{b \cdot b^2}{4a^2} \left[1 - \frac{yb}{a} \right]$$

$$F = \frac{b^3}{4a^2} \left[1 - \frac{3by}{2a} - 1 + \frac{yb}{a} \right]$$

$$= \frac{b^3}{4a^2} \left[\frac{-by}{2a} \right] = -\frac{b^4}{8a^3} y$$

$$T = 2\pi \sqrt{\frac{8a^3 m}{b^4}} = \frac{4\pi a}{b^2} \sqrt{2ma}$$

$$T = \frac{4\pi a}{b^2} \sqrt{2ma}$$

4. **Ans (16)**

$$\frac{1}{F_{\text{lens}}} = (1.5 - 1) \left[\frac{1}{-40} - \frac{1}{-20} \right] = \frac{1}{80}$$

$$F_{\ell} = 80 \text{ cm}$$

$$F_m = -\frac{20}{2} = -10 \text{ cm}$$

$$\frac{1}{F_{\text{eq}}} = \frac{1}{f_m} - \frac{2}{f_{\ell}} = \frac{1}{-10} - \frac{2}{80}$$

$$f_{\text{eq}} = -8 \text{ cm}$$

Hence object should be placed at $x = 16 \text{ cm}$.

i.e. at the centre of curvature.

5. **Ans (100)**

$$f_A = \frac{103}{100} f_0 \quad f_B = \frac{97}{100} f_0$$

$$f_A - f_B = 6$$

$$\frac{(103 - 97)}{100} \times f_0 = 6$$

$$\frac{6}{100} f_0 = 6$$

$$f_0 = 100 \text{ Hz}$$

6. **Ans (8)**

$$\frac{I_{\max}}{I_{\min}} = \left(\frac{\sqrt{I_1} + \sqrt{I_2}}{\sqrt{I_1} - \sqrt{I_2}} \right)^2 = \left(\frac{\sqrt{4} + \sqrt{1}}{\sqrt{4} - \sqrt{1}} \right)^2 = 9$$

and visibility $n = 0.8$

7. **Ans (3)**

$$\text{Angular width} = \frac{\lambda}{d} = 10^{-3} \text{ (given)}$$

$\therefore$ No. of fringes within 0.12 will be

$$n = \frac{0.12 \times 2\pi}{360 \times 10^{-3}} \cong [2.09]$$

$\therefore$ The number of bright spots will be three.

8. **Ans (30)**

$$V_B - V_A = - \int E_x dx = - [\text{Area under } E_x - x \text{ curve}]$$

$$V_B - 10 = -\frac{1}{2} \cdot 2 \cdot (-20) = 20$$

$$V_B = 30 \text{ V.}$$

9. **Ans (2)**

$$E = \frac{3KQ^2}{5R} \quad E_{\text{out}} = \frac{KQ^2}{2R}$$

$$E_{\text{in}} = \frac{KQ^2}{10R}$$

$$\text{Given } E_1 = E_2 = E_3 = \frac{KQ^2}{5R}$$

$$\text{therefore } \frac{KQ^2}{2R_2} = \frac{KQ^2}{5R} \Rightarrow R_2 = \frac{5R}{2}$$

$$\text{and } \frac{KQ^2}{2R_1} = \frac{2KQ^2}{5R} = R_1 = \frac{5R}{4}$$

$$\frac{R_2}{R_1} = 2$$

10. **Ans (100)**

Let intensity of sound be I and I' Loudness of sound initially

$$\beta_1 = 10 \log \left(\frac{I}{I_0} \right)$$

$$\text{Later } \beta_2 = 10 \log \left(\frac{I'}{I_0} \right)$$

$$\text{Given } \beta_2 - \beta_1 = 20$$

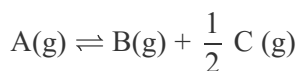
$$\therefore 20 = 10 \log \left(\frac{I'}{I} \right)$$

$$\therefore I' = 100 I$$

PART-2 : CHEMISTRY

SECTION-I

2. **Ans (B)**



$$\text{Initial : } P_i \quad 0 \quad 0$$

$$\text{At eq.: } P_i(1 - \alpha) \quad P_i \cdot \alpha \quad P_i \frac{\alpha}{2}$$

Now, equilibrium pressure (p),

$$P = P_i \times \left(1 + \frac{\alpha}{2} \right)$$

$$\therefore P_A = \left(\frac{1 - \alpha}{1 + \frac{\alpha}{2}} \right) P$$

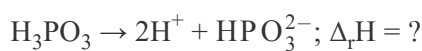
$$P_B = \left(\frac{\alpha}{1 + \frac{\alpha}{2}} \right) P$$

$$P_C = \left(\frac{\frac{\alpha}{2}}{1 + \frac{\alpha}{2}} \right) P$$

$$\therefore K = \frac{P_C^{\frac{1}{2}} \times P_B}{P_A}$$

$$K = \frac{\alpha^{\frac{3}{2}} p^{\frac{1}{2}}}{(2 + \alpha)^{\frac{1}{2}} (1 - \alpha)}$$

3. **Ans (B)**



$$\Delta_r H = -55.84 \times 2 = -111.68$$

$$-106.68 = \Delta_{\text{ion}} H - 55.84 \times 2$$

$$\Delta_{\text{ion}} H = 5 \text{ kJ/mol}$$

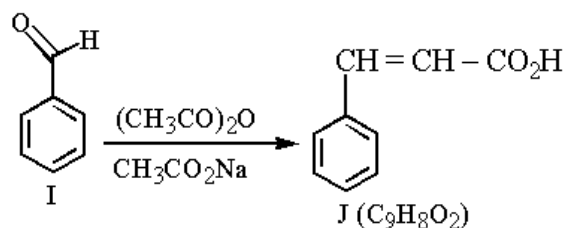
5. **Ans (A)**

For zero order reactions

$$\frac{t_1}{2} = \frac{a}{2k}$$

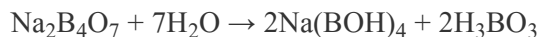
$$\frac{t_1}{2} \times a$$

12. **Ans (B)**



J gives effervescence on treatment with NaHCO_3 due to presence of $-\text{CO}_2\text{H}$ group and positive Baeyer's test due to presence of unsaturation.

14. **Ans (B)**



15. **Ans (C)**



16. **Ans (A)**

Reducing property of dioxide decrease from SO_2 to TeO_2

17. **Ans (C)**



18. **Ans (D)**

all are correct

19. **Ans (A)**

- (a) Both LiCl and MgCl_2 are soluble in ethanol
- (b) Li and Mg do not form superoxide
- (c) LiF has high lattice energy
- (d) Li_2O is least soluble in water than other alkali metal oxides

20. **Ans (A)**

Rate law for acid catalysed hydrolysis of ester is

$$\text{rate} = k [\text{H}^+] [\text{ester}]$$

$$\frac{[\text{rate}]_{\text{weak}}}{[\text{rate}]_{\text{strong}}} = \frac{1}{100} = \frac{[\text{H}^+]_{\text{weak}}}{[\text{H}^+]_{\text{strong}}} = \frac{[\text{H}^+]_{\text{weak}}}{1}$$

$$[\text{H}^+]_{\text{weak}} = 0.01$$



$$K_a = \frac{x^2}{C-x} = \frac{(0.01)^2}{1-0.01} \approx 10^{-4}$$

PART-2 : CHEMISTRY

SECTION-II

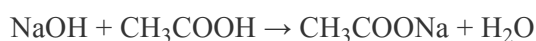
1. **Ans (6)**

i, iii, iv, v, vi, vii

3. **Ans (476)**

Moles of $\text{CH}_3\text{COOH} = 5$ m mole

moles of $\text{NaOH} = 2.5$ m mole



2.5 m mole 2.5 m mole

0 2.5 m mole 2.5 m mole

so buffer is formed

$$\text{pH} = \text{pK}_a + \log$$

$$\left(\frac{2.5/75}{2.5/75} \right) = \text{pK}_a$$

$$\text{pH} = 4.76$$

$$= 476 \times 10^{-2}$$

4. **Ans (3360)**

$$\Delta G = \Delta G^\circ + RT \ln Q$$

$$\Delta G^\circ = 2\Delta G_f^\circ(\text{NO}_2) - \Delta G_f^\circ(\text{N}_2\text{O}_4) = 0$$

$$\Delta G = 8 \times 300 \ln \left(\frac{2^2}{1} \right) = 3360 \text{ J}$$

5. **Ans (45)**

$$w = 2.5 \text{ g} \quad K_b = 0.52$$

$$w_{\text{solvent}} = 75 \text{ g} \quad M = \text{Mol. Wt. of solute}$$

$$T_B' = 373.535 \text{ K}$$

$$T_B^\circ = 373.15 \text{ K}$$

$$\Delta T_B = 0.385 = K_b \text{ molality}$$

$$0.385 = 0.52 \times \left(\frac{2.5}{M} \times \frac{1000}{75} \right)$$

$$M = 45 \text{ g mol}^{-1}$$

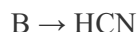
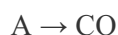
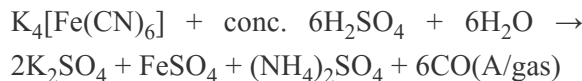
6. **Ans (6)**

i, ii, v, vi, vii, viii

8. **Ans (6)**



(excess) (x)



Total π bonds = 6.

PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B)**

$$h(f(g(x))) = h\left(f\left(\sqrt{x^2+1}\right)\right) = h(x^2)$$

$$\therefore x^2 \geq 0$$

$$h(x) = \begin{cases} 0, & x = 0 \\ x^2, & x \neq 0 \end{cases}$$

2. **Ans (D)**

$$2\vec{a} \cdot \vec{b} = -4\vec{a} \cdot \vec{b} + 3\vec{b}^2$$

$$\Rightarrow 2\vec{a} \cdot \vec{b} = \vec{b}^2$$

$$\text{Now } \vec{a} \cdot \vec{b} + \frac{4}{|\vec{b}|^2 + 2}$$

$$= \left(1 + \vec{a} \cdot \vec{b}\right) + \frac{4}{2(\vec{a} \cdot \vec{b} + 1)} - 1 \geq 2\sqrt{2} - 1$$

3. **Ans (A)**

Characteristic equation

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} 1-\lambda & 0 & 0 \\ 0 & 1-\lambda & 1 \\ 0 & -2 & 4-\lambda \end{vmatrix} = 0$$

$$\lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$$

$$A^3 - 6A^2 + 11A - 6I = 0 \quad \dots(i)$$

$$6A^{-1} = A^2 - 6A + 11I$$

4. **Ans (B)**

$$\vec{d} = \lambda (\vec{a} \times \vec{b})$$

$$1 = |\lambda| \cdot 1.1 \cdot \sin 30^\circ$$

$$\lambda = \pm 2$$

$$\pm \frac{\vec{d}}{2} = (\vec{a} \times \vec{b})$$

$$(\vec{a} \times \vec{b}) \times (\vec{c} \times \vec{d}) = \pm \frac{1}{2} \{ \vec{d} \times (\vec{c} \times \vec{d}) \}$$

$$= \pm \frac{1}{2} \{ \vec{c} - (\vec{c} \cdot \vec{d}) \vec{d} \}$$

$$= \pm \frac{\vec{c}}{2} = \frac{i-2j+2k}{6} \Rightarrow \vec{c} = \pm \left(\frac{i-2j+2k}{3} \right)$$

5. **Ans (A)**

$$\begin{vmatrix} x & 1 & 2 \\ f(x) & 3 & x^2 \\ 5x & 6 & 1 \end{vmatrix} = 0$$

$$x(3-6x^2) - 1(f(x)-5x^3) + 2(6f(x)-15x) = 0$$

$$3x - 6x^3 - f(x) + 5x^3 + 12f(x) - 30x = 0$$

$$11f(x) = x^3 + 27x$$

$$f(x) = \frac{x^3 + 27x}{11}$$

6. **Ans (D)**

$$(A) = \int_{-\pi}^{\pi} \underbrace{\cos 2x \cdot \cos 2^2x \cdot \cos 2^3x \cdot \cos 2^4x}_{f(x)} dx$$

$$I = 2 \int_0^{\pi} f(x) dx \quad [f(x) \text{ is even}]$$

$$I = 2.2 \int_0^{\pi/2} f(x) dx \quad \because f(\pi-x) = f(x)$$

$$I = 4 \int_0^{\pi/2} f(x) dx = 4I_1$$

$$\text{Now, } I_1 = \int_0^{\pi/2} f(x) dx$$

$$I_1 = -I_1 \quad \because f\left(\frac{\pi}{2} - x\right) = -f(x)$$

$$I_1 = 0$$

$$\Rightarrow I = 0$$

7. **Ans (A)**

$$\theta_1 + \theta_2 = \pi$$

$$\cos \theta_2 = \cos \left(\cos^{-1} \frac{4}{5} + \cos^{-1} \frac{1}{3} \right)$$

$$= \frac{4}{5} \times \frac{1}{3} - \frac{3}{5} \times \frac{\sqrt{8}}{3} = \frac{4 - 6\sqrt{2}}{15} < 0$$

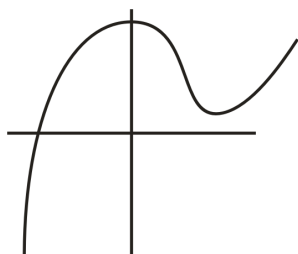
θ_2 is obtuse

θ_1 is acute

$$\theta_1 < \theta_2$$

8. **Ans (D)**

$$f(x) = 2x^3 - 3x^2 + 6$$



$$F'(x) = 6x^2 - 6x = 6x(x - 1)$$

$$F'(x) = 0$$

$$\text{At } x = 0, x = 1$$

Function must be one-one onto for

inverse to be exist domain $\in [1, \infty)$, $a \geq 1$

9. **Ans (A)**

$$I = \int e^x \left(\tan^{-1} x + \frac{2x}{(1+x^2)^2} \right) dx$$

$$= \int e^x \tan^{-1} x dx + \int e^x \frac{2x}{(1+x^2)^2} dx$$

$$= e^x \tan^{-1} x - \int \frac{e^x}{1+x^2} dx + 2 \int \frac{xe^x}{(1+x^2)^2} dx$$

$$= e^x \tan^{-1} x - \left[\frac{e^x}{1+x^2} - \int \frac{-e^x \cdot 2x dx}{(1+x^2)^2} \right] + 2 \int \frac{xe^x}{(1+x^2)^2} dx$$

$$= e^x \left(\tan^{-1} x - \frac{1}{1+x^2} \right) + C$$

$$f(0) = -1$$

$$C = 0$$

$$f(1) = e \left(\frac{\pi}{4} - \frac{1}{2} \right)$$

10. **Ans (A)**

$$\sqrt{x+y} + \sqrt{y-x} = C \quad \dots(i)$$

$$\frac{y+x - (y-x)}{\sqrt{x-y} - \sqrt{y-x}} = C$$

$$\sqrt{y+x} - \sqrt{y-x} = \frac{2x}{C} \quad \dots(ii)$$

Adding (i) and (ii)

$$2\sqrt{y+x} = C + \frac{2x}{C}$$

$$4(y+x) = C^2 + \frac{4x^2}{C^2} + 4x$$

$$4y = C^2 + \frac{4x^2}{C^2}$$

$$\frac{4dy}{dx} = \frac{4}{C^2} \cdot 2x$$

$$\frac{d^2y}{dx^2} = \frac{2}{C^2}$$

11. **Ans (A)**

$$I_n = \int \operatorname{cosec}^n 2x dx = \int \operatorname{cosec}^2 2x \operatorname{cosec}^{n-2} 2x dx$$

$$\Rightarrow I_n = -\frac{1}{2} \cot 2x \operatorname{cosec}^{n-2} 2x + \int \frac{1}{2} (\cot 2x) (n-2)$$

$$(-\operatorname{cosec}^{n-3} 2x \operatorname{cosec} 2x \cot 2x) 2 dx$$

$$\Rightarrow -\frac{1}{2} \cot 2x \operatorname{cosec}^{n-2} 2x - (n-2)$$

$$\int \cot^2 2x \operatorname{cosec}^{n-2} 2x dx$$

$$\Rightarrow I_n = -\frac{1}{2} \cot 2x \operatorname{cosec}^{n-2} 2x - (n-2)$$

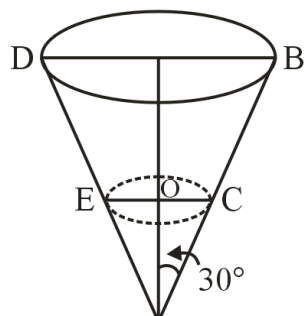
$$\int (\operatorname{cosec}^2 2x - 1) \operatorname{cosec}^{n-2} 2x dx$$

$$\Rightarrow (n-1) I_n = -\frac{1}{2} \cot 2x \operatorname{cosec}^{n-2} 2x + (n-2) I_{n-1}$$

$$\Rightarrow I_n = \frac{\cot 2x \operatorname{cosec}^{n-2} 2x}{-2(n-1)} + \frac{n-2}{n-1} I_{n-2}$$

12. Ans (C)

In $\triangle AOC$,



$$\tan 30^\circ = \frac{OC}{h}$$

$$OC = \frac{h}{\sqrt{3}}$$

$$V = \frac{1}{3}\pi \times \left(\frac{h^2}{3}\right) \times h = \frac{\pi h^3}{9}$$

$$\frac{dv}{dt} = \frac{\pi}{3} \times h^2 \frac{dh}{dt}$$

$$\frac{dv}{dt} = \frac{\pi h^2}{3} \left\{ \therefore \frac{dh}{dt} = 1 \text{ cm/s} \right\}$$

$$\frac{dv}{dt} = \frac{\pi \times (24)^2}{3} = 192\pi \text{ cm}^3/\text{s}.$$

13. Ans (B)

Equation of normal to $y = \tan x$ at $(\alpha, \tan \alpha)$

$$y - \tan \alpha = \frac{-1}{\sec^2 \alpha} (x - \alpha)$$

$$\tan \alpha \cdot \sec^2 \alpha = \left(2 + \frac{\pi}{4} - \alpha\right)$$

$$\tan \alpha (1 + \tan^2 \alpha) = \left(2 + \frac{\pi}{4} - \alpha\right)$$

$$\alpha = \frac{\pi}{4}$$

$\Rightarrow$ minimum distance = distance between

$$\left(\frac{\pi}{4}, 1\right) \text{ and } \left(2 + \frac{\pi}{4}, 0\right) - \text{radius} = \sqrt{5} - 1$$

14. Ans (A)

$$f(x) = 1 - x - x^3$$

$$f(f(x)) = 1 - f(x) - f^3(x)$$

$$f'(x) = -1 - 3x^2 \quad (\text{decreasing function})$$

$$\therefore f(x) < 1 - 5x$$

$$1 - x - x^3 < 1 - 5x$$

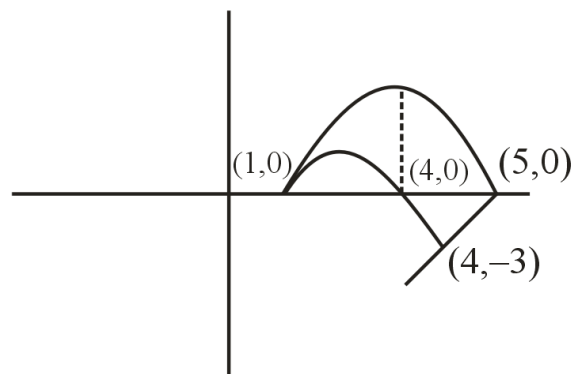
$$x^3 - 4x > 0$$

$$x(x^2 - 4) > 0$$

15. Ans (C)

$$A = \int_1^4 (-x^2 + 6x - 5 + x^2 - 4x + 3) dx$$

$$+ \int_4^5 (-x^2 + 6x - 5 - 3x + 15) dx$$



$$= \int_1^4 (2x - 2) dx + \int_4^5 (-x^2 + 3x + 10) dx$$

$$= [x^2 - 2x]_1^4 + \left[-\frac{x^3}{3} + \frac{3x^2}{2} + 10x \right]_4^5$$

$$= 9 + 10 + \frac{3}{2} \cdot 9 - \frac{1}{3} \cdot 61$$

$$= \frac{65}{2} - \frac{61}{3} = \frac{73}{6}$$

16. Ans (C)

$$\sum \frac{(-1)^r {}^{n+1}C_{r+1}}{(n+1)(r+2)(r+3)} = \sum \frac{(-1)^r {}^{n+2}C_{r+2}}{(n+1)(n+2)(r+3)}$$

$$= \sum \frac{(-1)^r {}^{n+3}C_{r+3}}{(n+1)(n+2)(n+3)} = \frac{1}{2(n+3)}$$

17. Ans (A)

$$\therefore \frac{2}{1!9!} + \frac{2}{3!7!} + \frac{1}{5!5!} = \frac{2^m}{n!}$$

$$\frac{1}{10!} \left\{ \frac{2 \times 10!}{1!9!} + \frac{2 \times 10!}{3!7!} + \frac{10!}{5!5!} \right\} = \frac{2^m}{n!}$$

$$\Rightarrow \frac{1}{10!} \{ 2^{10}C_1 + 2^{10}C_3 + {}^{10}C_5 \} = \frac{2^m}{n!}$$

$$\Rightarrow \frac{1}{10!} (2)^{10-1} = \frac{2^m}{n!}$$

$$\therefore m = 9 \text{ and } n = 10$$

18. Ans (A)

$$\therefore \lim_{x \rightarrow 0^+} (\sin x)^{\frac{1}{x}} = 0$$

$$\therefore \lim_{x \rightarrow 0^+} (\cos x)^{\frac{1}{x}} (1)^0 = e^{\lim_{x \rightarrow 0^+} \frac{\cos x - 1}{x}} = e^0 = 1$$

$$\therefore \lim_{x \rightarrow 0^+} (\operatorname{cosec} x)^x = \infty^0$$

$$= e^{\lim_{x \rightarrow 0^+} x \ln(\operatorname{cosec} x)} = e^{-\lim_{x \rightarrow 0^+} \frac{\ln(\sin x)}{\frac{1}{x}}}$$

$$= e^{-\lim_{x \rightarrow 0^+} \frac{\cot x}{x^2}} = e^0 = 1$$

19. Ans (B)

$$\frac{dy}{dx} = -2x(y-1)$$

$$\frac{dy}{dx} + 2xy = 2x$$

$$\text{I.F.} = e^{\int 2x dx} = e^{x^2}$$

$$y \cdot e^{x^2} = \int 2xe^{x^2} dx + c$$

$$y \cdot e^{x^2} = e^{x^2} + c \quad \dots(1)$$

$$\text{Given } y(0) = 1 \Rightarrow 1 \cdot 1 = 1 + c \Rightarrow c = 0$$

$$\therefore y \cdot e^{x^2} = e^{x^2} \quad (\text{from (1)})$$

$$y = 1$$

$$\therefore \lim_{x \rightarrow \infty} y(x) = \lim_{x \rightarrow \infty} (1) = 1$$

20. Ans (C)

$$2\vec{v} + \vec{v} \times (\hat{i} + 2\hat{j}) = 2\hat{i} + \hat{k} \quad \dots(1)$$

$$\text{Dot with } \hat{i} + 2\hat{j}$$

$$\Rightarrow \vec{v} \cdot (\hat{i} + 2\hat{j}) = 1$$

$$\Rightarrow |\vec{v} \cdot (\hat{i} + 2\hat{j})|^2 = 1$$

$$\Rightarrow |\vec{v}|^2 \cdot 5 \cdot \cos^2 \theta = 1$$

$$\Rightarrow |\vec{v}|^2 \cdot 5 \cdot \sin^2 \theta = 5|\vec{v}|^2 - 1 \quad \dots(2)$$

Taking mod in equation (1) using (2), we get

$$3|\vec{v}| = \sqrt{6}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (14)

$$A : x > x^2 - 3x + 3 \Rightarrow x^2 - 4x + 3 < 0$$

$$\Rightarrow x \in (1, 3)$$

$$\therefore f'(x) = 6x^2 - 6x + 12$$

$$= 6(x-2)(x+1)$$

$$\therefore f(x) \text{ is minimum at } x = 2$$

$$\therefore x = 2$$

$$\therefore f(2) = -14$$

2. Ans (12)

$$\therefore f(x) \text{ is surjective}$$

$$\therefore \text{minimum value of } ax^2 - 2px + 4 \text{ is } 1$$

$$\Rightarrow \frac{-\Delta}{4a} = 1 \Rightarrow 4p^2 = 12a$$

$$\Rightarrow p^2 = 3a \quad a, p \in \mathbb{N}$$

$$\therefore \text{for smallest } 2(a+p), a = 3, p = 3$$

$$\therefore 2(a+p) = 12$$

3. Ans (8)

Pre multiply by A

$$A^2 + A \cdot \text{adj } A = I_2$$

$$A^2 + |A| \cdot I_2 = I_2$$

$$A^2 = (1 - |A|)I_2$$

$$|A|^2 = (|A| - 1)^2$$

$$|A| = \pm(|A| - 1) \Rightarrow |A| = \frac{1}{2}$$

$$\text{Determinant } (2A^{-1}) = 8$$

4. Ans (44)

For A^{-1} exists $|A| \neq 0$

Case-I : Exactly one element is zero

$${}^5C_3 \times 4! = 240$$

Case-II : No element is zero

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

$$|A| = ad - bc = 0 \quad ad = bc = 6 \quad [8]$$

$$ad = bc = 12 \quad [8]$$

$$\Rightarrow {}^5C_4 \times 4! - 16 = 120 - 16$$

$$= 104$$

$$\text{Total} = 344$$

5. Ans (15)

$$\vec{r} \times \vec{b} = \vec{c} \times \vec{b} \Rightarrow (\vec{r} - \vec{c}) \times \vec{b} = \vec{0} \quad \dots(i)$$

$$\vec{r} \cdot \vec{b} = \vec{c} \cdot \vec{b} \Rightarrow (\vec{r} - \vec{c}) \cdot \vec{b} = 0 \quad \dots(ii)$$

from (i) and (ii) we get

$$\vec{r} - \vec{c} = \vec{0} \quad \left\{ \because \vec{b} \neq \vec{0} \right\}$$

$$\therefore \vec{a} \cdot \vec{r} = \vec{a} \cdot \vec{c} = (2\hat{i} + \hat{k}) \cdot (4\hat{i} + 3\hat{j} + 7\hat{k}) = 8 + 7 = 15$$

6. Ans (9)

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 = \begin{vmatrix} \vec{a} \cdot \vec{a} & \vec{a} \cdot \vec{b} & \vec{a} \cdot \vec{c} \\ \vec{b} \cdot \vec{a} & \vec{b} \cdot \vec{b} & \vec{b} \cdot \vec{c} \\ \vec{c} \cdot \vec{a} & \vec{c} \cdot \vec{b} & \vec{c} \cdot \vec{c} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & \cos \theta & \cos \theta \\ \cos \theta & 1 & \cos \theta \\ \cos \theta & \cos \theta & 1 \end{vmatrix}$$

$$= 2\cos^3 \theta - 3\cos^2 \theta + 1$$

$$= (2\cos \theta + 1)(1 - \cos \theta)^2$$

$$[\vec{a} \quad \vec{b} \quad \vec{c}]^2 \geq 0$$

$$\Rightarrow 1 + 2\cos \theta \geq 0$$

$$\Rightarrow \cos \theta \geq -\frac{1}{2}$$

$$\Rightarrow \theta \leq \frac{2\pi}{3}$$

7. Ans (12)

$$I = \underbrace{\int_0^2 (3x^2 - 6x + 4) \cos(x^3 - 3x^2 + 4x - 2) dx}_{I_1}$$

$$+ \underbrace{\int_0^2 (3x - 3) \cos(x^3 - 3x^2 + 4x - 2) dx}_{I_2}$$

Put $x^3 - 3x^2 + 4x - 2 = u$ in the first integral we get

$$I_2 = 3 \int_0^2 (x - 1) \cos((x - 1)^3 + 1(x - 1)) dx$$

$$I_1 = \int_{-2}^2 (\cos u) du = 2 \int_0^2 (\cos u) du \sin 2$$

Put $x - 1 = t$

$$\therefore I = I_1 + I_2 = 2 \sin 2$$

$$I_2 = 3 \int_{-1}^1 t \cos(t^3 + t) dt$$

$$\Rightarrow p = q = 2$$

$$= 0 \text{ (odd function)}$$

8. Ans (120)

$$\sum_{i=0}^{100} {}^{150}C_{150-i} \cdot {}^{350}C_{250+i} (150 - i)$$

$$150 \sum_{i=0}^{100} {}^{149}C_{149-i} \cdot {}^{350}C_{250+i}$$

$$150 {}^{499}C_{399} = k {}^{500}C_{100}$$

$$150 \left(\frac{500}{400} {}^{499}C_{399} \right) = k {}^{500}C_{100}$$

$$k = \frac{150 \cdot 400}{500}$$

$$k = 120$$

9. Ans (4)

$$\therefore 3^{4n} = 81^n = (1 + 80)^n = 1 + 80\lambda, \lambda \in \mathbb{N}$$

$$\therefore 3^{3^{4n}} = 3^{1+80\lambda} = 3 \cdot 3^{80\lambda} = 3 \cdot (9)^{40\lambda} = 3(10 - 1)^{40\lambda}$$

$$= 3(1 + 10\mu) = 3 + 30\mu$$

$$\therefore \text{Last digit of } 3^{3^{4n}} + 1 \text{ is } 4$$

10. Ans (1)

$$2x - y = 5$$

$$x - 2y = 4$$

$$x = 2 \quad y = -1$$

$$\Rightarrow x + y = 1$$